

Communication

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Chem. Mater., **Just Accepted Manuscript** • DOI: 10.1021/acs.chemmater.7b05108 • Publication Date (Web): 23 Jan 2018

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Correlation of structure and fast ion conductivity in the solid solution series $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$

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ABSTRACT: Solid electrolytes are the foundation of all-solid-state batteries (ASSB) and have the potential to provide improved safety and higher energy density than existing liquid battery systems. In the important search for new lithium ion conductors with fast ionic conductivity and good mechanical properties, thiophosphates are a particularly promising class of materials. The first experimental elucidation of new lithium ion conductors in the $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ (LZPS) solid solution whose existence had been predicted by theory is reported herein. Using neutron and synchrotron x-ray powder diffraction together with electrical impedance, and Raman studies, we resolve their crystalline nature and correlate this with ionic conductivity upon increasing the lithium/zinc ratio. We demonstrate that the materials exhibit high experimental ionic conductivities - up to $8 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$ - and show the nature of likely pathways for lithium ion conduction.

Rechargeable all-solid-state batteries (ASSBs) using inorganic solid electrolytes (SE) have become one of the most promising alternatives to conventional Li-ion batteries that use flammable organic liquid electrolytes. Inorganic SEs have the potential to enable batteries that are not only safer but have higher energy density, low temperature performance and longer cycle life¹⁻³. To realize the potential of ASSBs, new SE materials are sought that can adequately replace the current liquid electrolytes. These materials require fast lithium ion conduction ($>10^{-4} \text{ S}\cdot\text{cm}^{-1}$), a wide electrochemical stability window, and chemical compatibility with both the anode and cathode⁴⁻⁸. While a range of various lithium-ion conductors exist and have been studied⁹, sulfides show the most promise in terms of ionic conductivity and processability^{4,8}. Sulfides are very ductile, making them easily processed via cold-pressing. Notable sulfide type lithium-ion conductors that exhibit some of the highest ionic conductivities comparable to liquid electrolytes stem from the lithium thiophosphate family. These include the argyrodites,¹⁰ $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS: $12 \text{ mS}\cdot\text{cm}^{-1}$)¹¹ and its Si/Sn derivatives¹²⁻¹⁴, $\text{Li}_7\text{P}_3\text{S}_{11}$ ($17 \text{ mS}\cdot\text{cm}^{-1}$)^{15,16}, and most recently LGPS-type $\text{Li}_{9.54}\text{Si}_{1.74}\text{P}_{1.44}\text{S}_{11.7}\text{Cl}_{0.3}$ ($25 \text{ mS}\cdot\text{cm}^{-1}$)¹⁷ which exhibits the highest room temperature lithium ion conductivity to date.

Recently, Richards et al.¹⁸ used density functional theory and *ab initio* molecular dynamics to investigate the $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ (LZPS) system, which exhibits a body-centered cubic (bcc) sulfur sublattice. Theory suggests that this arrangement of sulfur atoms, which is also adopted by two other fast ion conductors ($\text{Li}_7\text{P}_3\text{S}_{11}$ and LGPS), is correlated to a low lithium migration barrier.¹⁹ The stoichiometric parent phase LiZnPS_4 and its optical properties have been previously reported but no ionic conductivity data was provided.^{20,21} Its structure consists of a frame-

work of PS_4^{3-} and ZnS_4^{6-} tetrahedra that house Li^+ ions, where Li fully occupies the 2b site and Zn fully occupies the 2a site (**Figure 1a**). In the $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ non-stoichiometric phases (**Figure 1b**) the additional Li shares the 2a site with Zn as we show below, with the excess Li distributing equally between the 2a and an additional site, 2d. This 2d site is close to the “interstitial” site that was not explicitly identified in the theoretical work.¹⁷ The stoichiometric phase was predicted to have extremely poor conductivity based on first-principles calculations ($1.81 \times 10^{-12} \text{ S}\cdot\text{cm}^{-1}$, $E_a = 1.07 \text{ eV}$), while the introduction of interstitial lithium defects is predicted to yield ionic conductivities greater than $5 \times 10^{-2} \text{ S}\cdot\text{cm}^{-1}$ (with $E_a > 0.17 \text{ eV}$) at room temperature up to a solid solution solubility limit of $x = 0.5$;¹⁸ i.e., $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$.

Here, we report the synthesis of the solid solution series, and the structure of the member within the solubility limit with the highest predicted conductivity, derived from combined X-ray and neutron powder diffraction analysis.

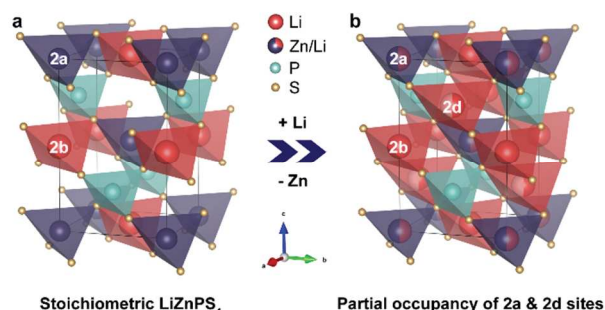


Figure 1. a) Crystal structure of stoichiometric LiZnPS_4 ; b) non-stoichiometric, showing addition of interstitial Li in the 2a site shared between Li and Zn and partial occupancy of Li

The highest value of x (i.e. of lithium excess) in a well crystallized sample was $x = 0.35$, corresponding to $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$. We unequivocally identify lithium occupation of the 2d site, which is charge compensated by substitution of a Zn^{2+} ion in the 2a site with Li^+ and is crucial for improving ionic transport. Experimental ionic conductivities (σ_i) measured by electrochemical impedance spectroscopy (EIS) show that the σ_i of the most highly conductive phases are in the range of $> 10^{-4} \text{ S}\cdot\text{cm}^{-1}$ at 25°C : a very significant improvement over LiZnPS_4 , albeit less than predicted. Phases with high values of x (> 0.5) were experimentally unattainable as relatively pure crystalline phases, however. Our observations demonstrate that the enhancement in ionic conductivity observed in the $x = 0.75$ phase is likely due to a glass-ceramic matrix effect rather than from its inherent crystalline structure.

$\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ compositions were prepared at $x = 0.25$, 0.5 and 0.75 using solid-state synthesis consisting of ball-milling and annealing. Their synchrotron x-ray diffraction (XRD) patterns are shown in **Figure 2**. The compositions contain small impurities of $\text{Li}_4\text{P}_2\text{S}_6$ and ZnS which form upon slight decomposition of the LZPS solid solutions because of their metastable nature. As the lithium content increases (up to $x = 0.75$), the formation of significant γ - Li_3PS_4 impurities becomes evident, in accord with the pseudo-binary phase diagram calculated by Richards *et al.*¹⁸.

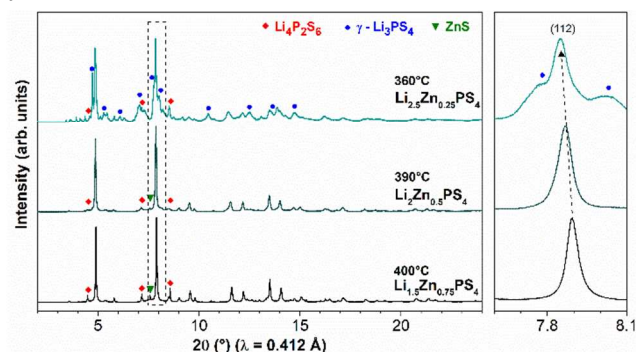


Figure 2. Synchrotron powder X-ray diffraction patterns (25°C) of $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ ($x = 0.25$), $\text{Li}_2\text{Zn}_{0.5}\text{PS}_4$ ($x = 0.5$), and $\text{Li}_{2.5}\text{Zn}_{0.75}\text{PS}_4$ ($x = 0.75$) with impurities of $\text{Li}_4\text{P}_2\text{S}_6$, ZnS , and $\gamma\text{-Li}_3\text{PS}_4$ as marked. Region with the dashed box is expanded on the right, indicating the shift in the (112) reflection ($I\bar{4}$ space group) towards higher angle. The corresponding synthesis temperatures are listed above the compositions.

The a and c lattice parameters were extracted by performing a Pawley full powder pattern fit of the synchrotron XRD patterns in the $I\bar{4}$ space group. The trends are shown in **Figure 3** (with exact values given in **Table S1**), indicating that the cell volume increases with increased lithium content ($293 \text{ \AA}^3$ to $301 \text{ \AA}^3$). The increase in cell volume stems from a significant increase in the a lattice parameter (with a small decrease in the c lattice parameter). Larger lattice volume often correlates to an improvement in ionic conductivity due to a reduced Li^+ ion migration barrier.^{19, 22} Since the targeted $\text{Li}_2\text{Zn}_{0.5}\text{PS}_4$ phase is expected to have the highest ionic conductivity within the solid solution solubility limit, its crystal structure was refined against combined synchrotron powder X-ray diffraction data ($\lambda = 0.412 \text{ \AA}$) and time-of-flight (TOF) neutron diffraction data.

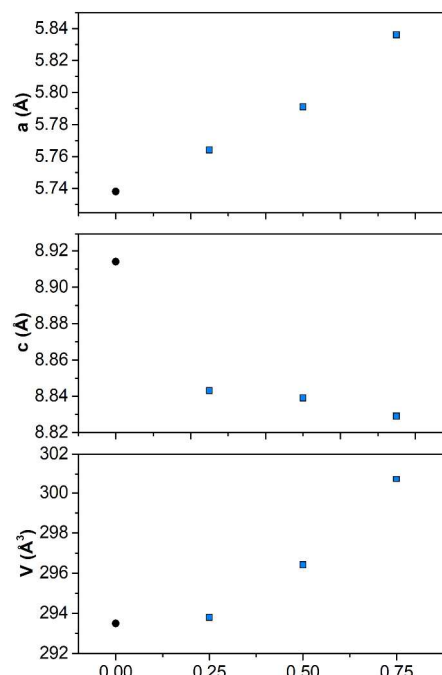


Figure 3. Lattice parameter and volume change in $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ for $x = 0, 0.25, 0.5$ and 0.75 . Solid black circles are from single crystal data taken from the literature¹⁹ and blue squares are experimental data from this work (values are listed in **Table S1**).

The reflections were assigned to a body-centered tetragonal cell (space group $I\bar{4}$, no. 82) with lattice parameters refined to $a = 5.7945(2) \text{ \AA}$, and $c = 8.8337(3) \text{ \AA}$. The crystallographic details are summarized in **Table 1a** (we note that bond lengths, **Table S2**, are typical for all environments) while the fit for the synchrotron and neutron data is shown in **Figure 4**.

The results of the refinement indicate that the actual lithium content of the crystalline material is $x \approx 0.35$; lower than that targeted (i.e. $x = 0.5$). The crystalline

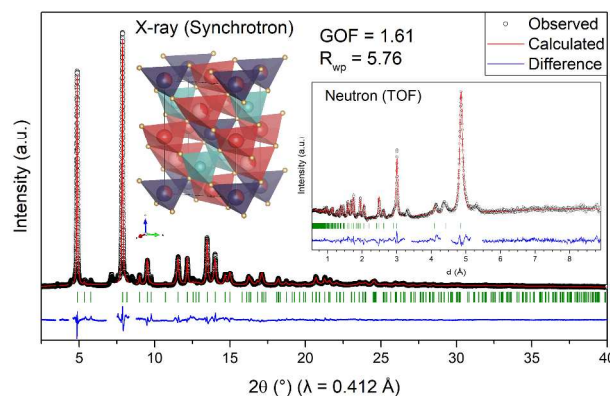


Figure 4. Combined Rietveld refinement of “ $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$ ” based on synchrotron x-ray diffraction and neutron diffraction data. Visual representation of the refined crystal structure (refined to a composition of $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$) is shown as an inset. Excluded regions correspond to impurity phases, mainly $\text{Li}_4\text{P}_2\text{S}_6$ which comprises $< 5\%$ of the total (see **Figure 2**).

Table 1a. Atomic coordinates, Wyckoff symbols, and isotropic displacement parameters for the targeted $\text{Li}_2\text{Zn}_{0.5}\text{PS}_4$ phase in the space group $I\bar{4}$ (no. 82).

Atom	Wyck. Pos.	x	y	z	SOF	B _{iso} (Å ²)
Li1	2d	0	0.5	0.75	0.345(9)	6
Li2	2b	0	0	0.5	1	8
Li3	2a	0	0	0	0.344(4)	1.49(7)
Zn3	2a	0	0	0	0.656(4)	1.49(7)
P1	2c	0	0.5	0.25	1	0.81(4)
S1	8g	0.7952(2)	0.2821(2)	0.1220(2)	1	1.71(4)

$\text{Li}_{1.687}\text{Zn}_{0.656}\text{PS}_4$ (i.e., $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$) phase thus exists in an amorphous Li-rich thiophosphate matrix. The degree of crystallinity was estimated to be 69% using Si as an internal reference (**Figure S1**). A proposed mixture of [77% $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ + 23% $\alpha\text{-Li}_3\text{PS}_4$] accounts for the overall “ $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$ ” target. The 8% discrepancy may be attributed to the impurity phases (Fig. 2). The refined structure of $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ shows that the 2a sites are partially occupied by Li (34%) and Zn (66%) while the 2d site has partial occupancy of Li (34%) as visually represented in the inset in **Figure 4**. Owing to the well-known difficulty in simultaneously refining thermal parameters and occupancies, the thermal parameters for the 2d and 2b site were fixed to average values from the literature.^{11,17,23,24} This resulted in convergence to site occupancies that yielded perfect charge balance of the compound and an equal distribution of excess Li on the 2d and 2a sites as expected. The results are in excellent accord with *ab initio* molecular dynamics studies that predict significant lithium ion density on an “interstitial site”,¹⁷ which is similar to the 2d Wyckoff site we identify.

While $\text{Li}_{2.5}\text{Zn}_{0.25}\text{PS}_4$ ($x = 0.75$) was projected by theory to have the highest conductivity ($1.14 \times 10^{-1} \text{ S}\cdot\text{cm}^{-1}$),¹⁷ its structure could not be refined owing to the low quality of the X-ray and neutron data for this poorly crystallized compound. This is in accord with it lying just out of the predicted solubility limit.¹⁷ The structure of “ $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ ” was refined from the synchrotron X-ray data owing to its low lithium content (and a paucity of neutron time); and these results are shown in **Table 1b**. The fit is shown in **Figure 5**; refined bond lengths (**Table S2**) correspond to typical values for all environments. The lattice

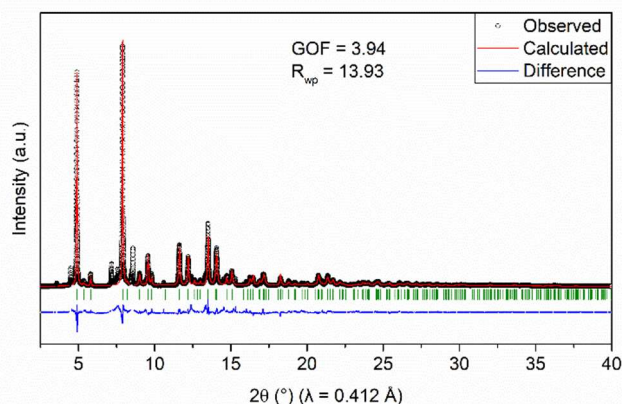
Table 1b. Atomic coordinates, Wyckoff symbols, and isotropic displacement parameters for the targeted $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ phase in the space group $I\bar{4}$ (no. 82)

Atom	Wyck. Pos.	x	y	z	SOF	B _{iso} (Å ²)
Li1	2d	0	0.5	0.75	0.097	6
Li2	2b	0	0	0.5	1	8
Li3	2a	0	0	0	0.097	1.30
Zn3	2a	0	0	0	0.903(4)	1.30 (6)
P1	2c	0	0.5	0.25	1	2.3(1)
S1	8g	0.7865(2)	0.2709(3)	0.1294(2)	1	0.89(4)

parameters refined to $a = 5.7641(1) \text{ \AA}$, and $c = 8.8438(2) \text{ \AA}$. Since lithium can not be accurately refined from X-ray data, lithium occupation in the 2d and 2a site was fixed to be equal, as determined by the $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ formulation. The Zn occupancy was refined and the vacant occupancy was assumed to be lithium given that the 2a site occupancy is constrained to 1. The refined composition of $\text{Li}_{1.2}\text{Zn}_{0.9}\text{PS}_4$ (with a 70% degree of crystallinity determined by standard addition) shows that the target composition was also not achieved in this case. This may be due to slow kinetics of the reaction that prevent achievement of the targeted lithium contents, despite the employment of nano-sized ZnS in the synthesis (see Supporting Information). However, synthesis at higher temperature or longer times is limited by thermodynamics: due to the metastability of the substoichiometric phases, the LZPS structure partially decomposes to $\text{Li}_4\text{P}_2\text{S}_6$. The formation of such impurity phases can be prevented by heating at lower temperatures; however, this comes at the cost of reduced degree of crystallinity.

The importance of excess phonon free energy in stabilizing defect-type ionic conductors with highly mobile Li^+ ion lattices has been previously invoked, for example, in $\text{Li}_7\text{P}_3\text{S}_{11}$, and especially in $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$.¹⁷ We note that the limiting composition $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ ($x = 0.35$) is approximately halfway between the predicted solubility limit for the solid solution series without phonon contributions ($x = 0.15$) and with phonon contributions ($x = 0.5$).¹⁸ Clearly $\text{Li}_{1.2}\text{Zn}_{0.9}\text{PS}_4$ ($x = 0.1$) falls within the regime where phonon contributions are not necessary to stabilize the material. All experimental attempts to increase the degree of lithium substitution to attain $x = 0.5$ were unsuccessful - irrespective of temperature, time and processing conditions - and resulted in significant levels of impurity phases. These results confirm that phonon contributions do play a vital role in stabilizing these defect structures but likely to a lower degree than presumed.

Partial occupancy of Li in the 2a and 2d site in the LZPS structure should facilitate the ion conduction by creating low energy percolating pathways through the structure. The Li^+ ion conductivity of the various LZPS compositions was studied using electrochemical impedance spectroscopy (EIS) on pressed pellets at variable temperatures. The impedance spectra in Nyquist form showed classic semicircles at high frequency, and a linear Warburg component in the low-frequency range (**Figure 6a**). Fitting the data (see inset for equivalent circuit and **Table S3** for fit parameters) resulted in experimental conductivities at 25°C plotted in **Figure 6b** as a function of lithium content.

**Figure 5.** Rietveld refinement of “ $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ ” synchrotron x-ray diffraction data only. Excluded regions correspond to impurity phases, mainly $\text{Li}_4\text{P}_2\text{S}_6$, which comprises <10% of the total (see **Figure 2**).

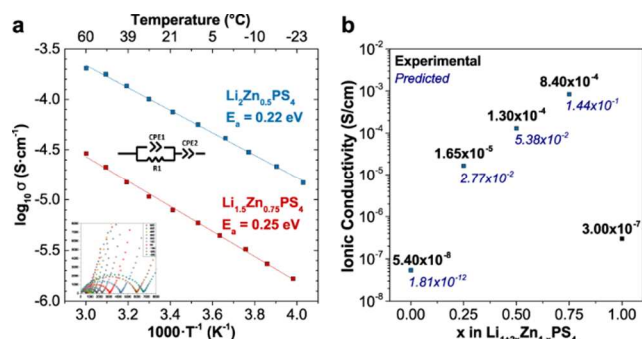


Figure 6. a) Activation energy plot of $\text{Li}_2\text{Zn}_{0.5}\text{PS}_4$ and $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ along with an inset of the Nyquist plots at each temperature plus the equivalent circuit used to fit the impedance data and b) Li^+ ion conductivity at 25°C versus x in $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$. Predicted values are taken from ref. 18.

The conductivities range from $5.40 \times 10^{-8} \text{ S}\cdot\text{cm}^{-1}$ at $x = 0$ to $8.4 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$ at $x = 0.75$. The conductivity of the refined phase $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ was $1.30 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$, which is lower than predicted for either $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$ ($5.38 \times 10^{-2} \text{ S}\cdot\text{cm}^{-1}$) or $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ ($2.77 \times 10^{-2} \text{ S}\cdot\text{cm}^{-1}$) as shown in **Figure 6**. The amorphous content may be responsible, although temperature-dependent conductivity measurements exhibit Arrhenius behavior with a low activation energy of 0.22 eV. This value is close to the range predicted by MD simulations for the defect phases,¹⁸ namely 0.165 eV ($\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$) and 0.181 eV ($\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$). While the possible amorphous phase (i.e., $\alpha\text{-Li}_3\text{PS}_4$; $\sigma_i = 2.8 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$)²⁵ may contribute to the overall conductivity, its much lower reported activation energy of 0.4 eV suggests that the σ_i of $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ dominates. Similarly, the refined phase $\text{Li}_{1.2}\text{Zn}_{0.9}\text{PS}_4$ has a room temperature ionic conductivity of $1.65 \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$, which is lower than the predicted $\text{Li}_{1.25}\text{Zn}_{0.875}\text{PS}_4$ phase ($3.44 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$), but the activation energy is 0.25 eV, quite close to theory (0.252 eV¹⁸).

The highest ionic conductivity of $8.4 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$ was achieved when the $x = 0.75$ glassy precursor was heated at lower temperatures (230°C) to form a very poorly crystallized material. An SEM image of its featureless morphology is shown in **Figure S2**. Heating above that temperature results in a lower ionic conductivity of $2.4 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$, and clear phase separation of the LZPS solid solution and $\gamma\text{-Li}_3\text{PS}_4$. The high conductivity in the glassy phase formed at low temperature is exclusive to the $x = 0.75$ compound (**Table S3**). Postulating that the improvement in ionic conductivity arises from the glass-ceramic nature of the material, we probed the local structure of $\text{Li}_{2.5}\text{Zn}_{0.25}\text{PS}_4$ via Raman spectroscopy, and compared it to the other two phases. (**Figure S3**). While $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$ and $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ show predominantly PS_4^{3-} moieties characteristic of the crystalline structure (along with $\text{Li}_4\text{P}_2\text{S}_6$ impurity), the more highly conductive $\text{Li}_{2.5}\text{Zn}_{0.25}\text{PS}_4$ synthesized at low-temperature contains an additional peak in the Raman spectrum not present in the other two compositions. It corresponds to $\text{P}_2\text{S}_7^{4-}$ units; inherent building blocks in the crystalline fast lithium ion conductor $\text{Li}_7\text{P}_3\text{S}_{11}$.¹⁶ They exist in its amorphous precursor (70:30 Li_2S - P_2S_5 glass), which is known to have very high conductivities of $>10^{-4} \text{ S}\cdot\text{cm}^{-1}$ before annealing²⁵. This leads us to propose that the glassy matrix plays a significant role in the high ionic conductivity of

" $\text{Li}_{2.5}\text{Zn}_{0.25}\text{PS}_4$ ", rather than inherently enhanced pathways in the structure.

In summary, our results provide an understanding of ion transport in the $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ solid solution series by defining the complex relationship between lithium site occupation/position and ionic conductivity. *Ab initio* computational techniques provide inspiration and guidance in the search for new solid-state ion conductors. Here, we have compared the results of introducing interstitial lithium defects in the solid solution regime $\text{Li}_{1+2x}\text{Zn}_{1-x}\text{PS}_4$ predicted by theory, with that of experiment. The stability of the defect phases owing to phonon contributions may be somewhat overestimated by theory, as all phases are found to be highly metastable. Such metastability results in difficulty in overcoming the kinetics necessary for mass transport in order to access the pure phases at the low temperatures required for their synthesis. Nonetheless, phases close to the targeted $\text{Li}_{2.0}\text{Zn}_{0.5}\text{PS}_4$ and $\text{Li}_{1.5}\text{Zn}_{0.75}\text{PS}_4$ were successfully synthesized. Their structures were elucidated via refinement of diffraction data. The obtained phases, $\text{Li}_{1.7}\text{Zn}_{0.65}\text{PS}_4$ and $\text{Li}_{1.2}\text{Zn}_{0.9}\text{PS}_4$ respectively, demonstrated ionic conductivities of $1.30 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$ and $1.65 \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$. These values are more than four orders of magnitude higher compared to that of the parent LiZnPS_4 phase, owing to occupation of the 2d interstitial site in the lattice as predicted by theory. This represents a good demonstration of how information from *ab initio* molecular dynamics can augment experiment, similar to $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, where early reports using powder diffraction alone could not locate a lithium site important to 3D mobility, but a combination of theory and single crystal diffraction enabled full structure resolution.^{11,26–28} While our experimental lithium ion conductivities were lower than predicted, this can be ascribed to a combination of not fully accessing the targeted lithium defect concentrations (owing to the limitations of kinetics and thermodynamics), and the accompanying reality of the presence of amorphous and potentially poorly conductive materials. The fact that we can synthesize solid solutions close to the predicted superionic phases is encouraging, however, and suggests that alternate highly non-equilibrium synthetic routes may be required to access pure phases. This should provide further inspiration for future endeavors. Moreover, LZPS compositions at high lithium contents – $\text{Li}_{2.5}\text{Zn}_{0.25}\text{PS}_4$ – exhibit conductivities up to $8 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$, showing that a glassy matrix can play a key role in governing fast lithium ion conduction.

ASSOCIATED CONTENT

Supporting Information.

This material is available free of charge via the Internet at <http://pubs.acs.org>.

Experimental details, lattice parameters for different compositions, bond lengths, SEM images and Raman spectra.

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ACKNOWLEDGEMENTS

This research was supported by the BASF International Scientific Network for Electrochemistry and Batteries. L.F.N. also

thanks NSERC for generous support via their Canada Research Chair, and Discovery Grant programs.

Use of the Advanced Photon Source at Argonne National Laboratory was supported by the U. S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. DE-AC02-06CH11357. Some of the data were collected as part of the 2017 Modern Methods in Rietveld Refinement and Structural Analysis School with support from its sponsors. We thank Saul H. Lapidus for his assistance with the synchrotron powder diffraction data collection at 11-BM.

Neutron diffraction measurements at the POWGEN and VULCAN instrument at ORNL's Spallation Neutron Source were sponsored by the Scientific User Facilities Division, Office of Basic Energy Sciences, US Department of Energy. We thank Yan Chen for his assistance with the neutron powder diffraction data collection at VULCAN.

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